David Oshidero — MEng AMIMechE MINCOSE

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PROFILE

Experienced Systems and Software Engineer specialising in Model-Based Systems & Software Engineering (MBS&SE) and Enterprise Architecture, with cross-sector experience across Defence, Aerospace, Nuclear, and Energy. Proven ability to develop mission and capability architectures using SysML, ArchiMate, and NAFv4, whilst delivering software solutions in Java, Python, and JavaScript. Adept at bridging enterprise, systems & software architecture with engineering delivery, engaging stakeholders at programme level and leading technical teams to produce consistent, traceable, and high-quality designs.

TECHNICAL SKILLS

- **Management:** Stakeholder Management, Resource Management, Risk Management, Bidding, Agile Approach, Waterfall Approach, Stakeholder Communication, Strategic Planning.
- **Programming Languages (PL):** MATLAB, VBA, Python, Java, JavaScript (ES6).
- **Programming Language Libraries:** JavaFX, Cameo Systems Modeller (Java), Sparx EA – SpiderMonkey.
- **Modelling Languages:** SysML, UML.
- **Frameworks:** MOFLT (Airbus), MagicGrid, TOGAF, NAF,.ROBOT, BPMN, Six Sigma, Systems Engineering V-Model, Software Development Lifecycle (SDLC), Docker, React, React-Native PyTorch.
- **Engineering Software:** DOORS, Sparx EA (Modelling & Scripting), Cameo Systems Modeller (Modelling & Plugins), ArchiMate, SOLIDWORKS, Ansys, Simulink, CISCO Packet Tracer, WireShark.
- **Cloud & Virtualization:** AWS, Google Cloud VMs, VMWare Workstation 10, Proxmox, SAP
- **Other Software & Tools:** Microsoft Office Packages, Confluence.

WORK EXPERIENCE

AtkinsRealis - Senior Systems Engineer (Enterprise Architecture) *June 2025 - Present*

- Led the development of Mission Engineering models using NAFv4 and ArchiMate, applying both Navy and MOD Mission Engineering frameworks to align operational capability needs with system development; spanning strategic intent through to physical system realisation.
- Extended mission modelling capability into SysML, bridging the gap between enterprise-level operational architectures and systems engineering design models to improve cross-discipline traceability.
- Developed Enterprise Architectures across business, application, and technology layers for MOD programmes, producing capability models, application landscape views, and technology roadmaps to support investment decisions and capability planning.
- Engaged stakeholders across engineering and programme teams to translate operational needs into structured architecture views, ensuring traceability from strategic intent to solution design.
- Delivered training and internal guidance on Sparx EA, SysML, and ArchiMate, building team competency and driving consistency and quality across architecture models.

Capgemini - Senior Software & Systems Engineer (MBS&SE) *November 2022 – June 2025*

- Successfully implemented MBSE into the ITER Project with Fusion4Energy, developing a pilot model using MagicGrid in Cameo to create a digital twin of the Tokamak Building, and delivering training to build MBSE competency within Fusion4Energy and raise executive awareness of its implementation value.
- Conducted requirement change analysis for a defence project, including performing impact assessments to identify risks and inform risk management decisions.
- Contributed to the Airbus NextWing and OneHeart initiatives, leveraging digital tools and semantic integration to accelerate wing design, and developing architecture models using AIRBUS' proprietary MOFLT language across functional, logical, and technical layers (ontological levels 1–4) to support system design realisation.
- Designed and implemented an end-to-end Java interface between Cameo and Delmia, reducing manual effort in model updates and accelerating data transfer between tools.
- Developed a Java plugin with XML marshalling against a custom ontology, improving data handling and increasing processing speed by 15% between initial and final versions, managed via Kanban.
- Led the development of an ontology-driven adapter between Sparx EA and a Neo4j data warehouse for XML data transfer using JavaScript, managing a team of 4 engineers using an Agile approach; overseeing sprint planning, workload distribution, and delivery, producing tool documentation, and architecting supporting test models in Sparx EA using NAF.
- Designed and implemented a Neo4j graph database as a central data warehouse, extracting and querying model data via MySQL.
- Developed and delivered MBSE training packs for MBDA, training internal teams and external clients, improving MBSE competency across 5+ engineering departments.

Jacobs - Systems/Enterprise Architect *February 2022 - November 2022*

- Engineering consultant contracted as a Systems/Enterprise Architect to Network Rail for the Transpennine Route Upgrade project in the railway industry.

- Developed Enterprise Architecture for whole project, individual systems and at various key outputs in the design process as well as a WBS for the project based on business and systems requirements available, using TraK and TOGAF.
    - Established an Architecture Plan, baselining the "As-Is" and "To-Be" architectures to guide system evolution.
    - Conducted SWOT and PESTLE analyses to assess business strengths, weaknesses, and external factors impacting project success, identifying key risks and opportunities that informed strategic time and integration decisions and mitigated project delays. Developed business architecture including architecture roadmaps that aligned with Network Rail's business requirements, ensuring consistency across project objectives.
    - Addressed interfacing risks and applied mitigation strategies between components in architecture diagrams, reducing potential issues.
- Ensured comprehensive system definitions for architecture designs, enhancing the accessibility of essential information in turn improving team collaboration.

Alten Engineering - Software & Systems Engineer *August 2021 - February 2022*

- Engineering consultant contracted to Rolls-Royce as a Systems and Software Verification Engineer for the ECOSIStem project, aimed at streamlining code testing and development for engine maintenance and enhancement.
- Utilized Python and .ROBOT framework to develop and analyse test cases as well as creating integration tests, ensuring easy testing of new code implementations.
- Reviewed, revised and verified test cases, improving quality and ensuring compliance, facilitating necessary reworks for approval.
- Created a knowledge-sharing database, improving onboarding efficiency for new consultants and enhancing collaborative information exchange, reducing onboarding time and improving knowledge retention.

Rolls-Royce - Performance Engineer/Development Engineer *June 2018 - August 2019*

- Reformed and implemented a MATLAB trending tool that tracked the deterioration rate of engines, incorporating advanced analytical processes (using VBA for data storage), and validated the tool, significantly enhancing data accuracy and processing time by 25%. Allowed for more accurate time resource planning for servicing.
- Managed two projects, where responsibilities included tracking project charters, managing resources, and ensuring deliverables were met whilst identifying and managing risks.
- Wrote Pass-off trending reports and conducted analysis for Rolls-Royce-manufactured jet engines.
- Defined and sequenced verification strategies aligning with customer, business, and aviation authority (EASA) requirements for the Electric Propulsion Unit system, aiming to achieve 100% compliance with EASA CS-Es.
- Took ownership of delivering key sub-systems including Fan Systems, Structures & Flow paths, and Bypass & Exhausts.

EDUCATION

Aerospace Engineering (MEng) - Swansea University *1st Achieved*

- Extra Qualifications: Python for Data Science courses (via DataCamp)
- Awarded 'Outstanding Achievement Award' in Year 3

Networking and Cybersecurity (Level 3 Diploma) *Awarded by Gateway Qualifications*

Certifications

INCOSE ASEP, SAP Technology Consultant, AWS Fundamentals, AWS Cloud Technology Consultant, Gas Turbines Performance, Engine Lubrication Systems, MATLAB Fundamentals

PERSONAL ACHIEVEMENTS & VOLUNTEERING

- Volunteering & STEM: Involved with helping to build the new garden in the MS Therapy Centre in Bristol, a great opportunity to give back to the local community.
- Working at Rolls-Royce has also allowed me to get involved in STEM projects such as the Flying Start Challenge, a group challenge for local school children, which is a fulfilling activity as it allows inspiring engineers of tomorrow.
- Associate member of both IMechE and INCOSE.